MMengineering

Finance

ETHOS System Guide

ETHOS ERP | Finance & Accounting
Version 1.0 | March 2026

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1. Overview

The ETHOS Finance module is a full double-entry accounting system built specifically for MM Engineered Solutions. It replaces the need for standalone accounting software by providing integrated ledgers, bank management, VAT compliance, and construction contract accounting — all connected to the wider ETHOS ERP.

Every financial transaction in the system creates a balanced journal entry where total debits always equal total credits. The module is designed for HMRC Making Tax Digital (MTD) compliance and follows UK GAAP conventions.

Core Principles

- Double-entry bookkeeping — every transaction debits one or more accounts and credits one or more accounts
- All amounts use Decimal precision to avoid floating-point rounding errors
- Auto-journal generation — posting a sales invoice, purchase invoice, payment, or bank transfer automatically creates the correct double-entry journal
- Period-based accounting — transactions must be posted to an open accounting period
- Audit trail — journals cannot be deleted, only reversed (creating a counter-entry)

Navigation

The Finance module uses a dedicated sidebar with collapsible sections:

- Overview — Dashboard, Invoicing, Job Costing, Nominal Codes
- Sales — Sales Ledger, Credit Control
- Purchases — Purchase Invoices, Enquiries
- Banking — Bank & Payments, Bank Rules
- Accounting — Chart of Accounts, Journal Entries, Contracts, Fixed Assets, Prepayments, Recurring Entries, Cost Centres
- Tax & Compliance — VAT Returns, Budgets, Reports
- Administration — Periods, Year-End, Import Data, Sage Export, Settings

2. Chart of Accounts

The Chart of Accounts is the foundation of the accounting system. Every nominal account has a code, name, type, and normal balance direction. Accounts support a parent/child hierarchy for structured reporting.

Account Types

Type	Normal Balance	Description
Asset	Debit	What the business owns (bank, debtors, stock, fixed assets)
Liability	Credit	What the business owes (creditors, VAT, loans)
Equity	Credit	Owner's capital, retained earnings, share capital
Revenue	Credit	Income from sales, contracts, other sources
Expense	Debit	Costs incurred (materials, labour, overheads)

Key System Accounts

The following well-known accounts are used by auto-journal functions and must not be renamed or deleted:

Code	Name	Purpose

1100	Trade Debtors	Sales ledger control — total owed by customers
1200	Current Account	Default bank GL account
1400	VAT Input	VAT reclaimable on purchases
2100	Trade Creditors	Purchase ledger control — total owed to suppliers
2200	VAT Output	VAT collected on sales
2302	CIS Deductions	Construction Industry Scheme deductions payable to HMRC
2400	Retention Held	Retention held on construction contracts
3100	Retained Earnings	Cumulative profit/loss carried forward at year-end

Working with Accounts

- **Add Account —** Finance > Chart of Accounts > Add Account. Enter code, name, type, sub-type, and optional parent account.
- **Hierarchy —** Accounts can be nested under parent accounts. Click the expand arrow to view children. Use this for grouping (e.g. 4000 Sales under 4xxx Revenue).
- **Filter by Type —** Use the tab bar (All, Assets, Liabilities, Equity, Revenue, Expenses) to quickly filter the list.
- **Search —** Search by code or name using the search bar.
- **Activate / Deactivate —** Toggle the Active/Inactive switch. Inactive accounts cannot receive new postings but retain their history.

> System accounts (1100, 1200, 2100, 2200, etc.) should never be deactivated — they are required by the auto-journal engine.

3. Journal Entries

Journal entries are the atomic unit of the accounting system. Every financial event — invoices, payments, transfers, depreciation — is recorded as a journal entry with balanced debit and credit lines.

Journal Statuses

Status	Meaning
JOURNAL_DRAFT	Entry created but not posted. Can be edited or deleted.
POSTED	Entry posted to the ledger. Cannot be edited — only reversed.
REVERSED	A counter-entry has been created to cancel this journal.

Creating a Manual Journal

Navigate to Finance > Journal Entries > New Journal.

- Enter a date, description, and optional reference
- Add two or more lines — each with an account, description, and either a debit or credit amount
- The system validates that total debits equal total credits before allowing save
- Each line can optionally have a VAT code, project, or cost centre tag

> A journal line must have either a debit OR a credit — never both, never neither. The validation engine enforces this rule.

Posting a Journal

From the journals list, click the green tick icon on any draft entry. Posting is irreversible — the entry locks and updates all affected account balances.

Reversing a Journal

Click the red undo icon on any posted entry. The system creates a new mirror journal that swaps debits and credits, effectively cancelling the original. Both entries remain in the audit trail.

Journal Sources

Every journal records its source for audit purposes:

Source	Created By
MANUAL	User-created journal entry
SALES_INVOICE	Auto-journal when a sales invoice is posted
PURCHASE_INVOICE	Auto-journal when a purchase invoice is posted
CREDIT_NOTE	Auto-journal when a credit note is posted
BANK_RECEIPT	Auto-journal when a customer payment is received
BANK_PAYMENT	Auto-journal when a supplier payment is made
BANK_TRANSFER	Auto-journal when an inter-bank transfer is made
CONSTRUCTION_APPLICATION	Auto-journal when a contract application is certified
YEAR_END	Auto-journal for year-end P&L to retained earnings transfer

Auto-Journal Details

The auto-journal engine (src/lib/finance/auto-journal.ts) creates journals automatically when operational actions occur. Key patterns:

Sales Invoice Posted

- DR Trade Debtors (1100) — gross amount
- CR Revenue account — net amount
- CR VAT Output (2200) — VAT amount

Purchase Invoice Posted

- DR Expense / Cost account — net amount
- DR VAT Input (1400) — VAT amount
- CR Trade Creditors (2100) — gross amount

Customer Payment Received

- DR Bank account — payment amount
- CR Trade Debtors (1100) — payment amount

Supplier Payment Made

- DR Trade Creditors (2100) — payment amount
- CR Bank account — payment amount

Bank Transfer

- DR Destination bank GL account — transfer amount
- CR Source bank GL account — transfer amount

Credit Note

- DR Revenue account — net amount (reversal)
- DR VAT Output (2200) — VAT amount (reversal)

- CR Trade Debtors (1100) — gross amount

4. Sales Ledger

The Sales Ledger manages all customer invoicing, credit notes, and receivables tracking. Navigate to Finance > Sales Ledger.

Sales Ledger Dashboard

The landing page shows four summary cards:

- Total Outstanding — all unpaid balances
- Current (0-30 days) — invoices within normal payment terms
- Overdue 30-60 days — amber warning zone
- Overdue 60+ days — red warning zone

Invoices

The Invoices tab lists all sales invoices with columns: Invoice #, Customer, Date, Due Date, Subtotal, VAT, Total, Paid, Outstanding, and Status.

- **New Invoice —** Click New Invoice to create. Select a customer, add line items with descriptions, quantities, unit prices, VAT codes, and revenue accounts.

Invoice numbering follows the sequence INV-XXXXXX, auto-generated by the sequence counter.

Invoice Statuses

Status	Meaning
ACC_DRAFT	Invoice created, not yet posted to the ledger
ACC_APPROVED	Approved but not yet posted
ACC_POSTED	Posted — auto-journal created (DR Debtors, CR Revenue, CR VAT)
ACC_PARTIALLY_PAID	Some payment received, balance still outstanding
ACC_PAID	Fully paid — outstanding amount is zero
ACC_CANCELLED	Invoice voided

Credit Notes

Credit notes reverse all or part of a sales invoice. They follow the same workflow but create a reversal auto-journal (DR Revenue, CR Trade Debtors). Numbering uses the CN-XXXXXX sequence.

Customers

The Customers tab lists all customer accounts with code, name, contact details, outstanding balance, and active status. Each customer has a unique code (CUST-XXXXXX).

Credit Control

Navigate to Finance > Credit Control for a dedicated debt-chasing view.

- Summary cards: Total Overdue, 30+ days, 60+ days, 90+ days
- Customer list sorted by days overdue or outstanding amount
- Each customer shows contact details, oldest invoice date, number of overdue invoices, credit limit
- Chasing actions: Reminder 1, Reminder 2, Reminder 3, Final Demand, Phone Call, Account on Hold, Legal Action, Write Off
- Next follow-up date tracking — overdue follow-ups highlighted in red

> Click any customer row to open their detailed credit control page with full invoice breakdown and action history.

Aged Debtors Report

Finance > Sales > Aged Debtors provides a breakdown of all outstanding invoices grouped into ageing bands: Current, 30 days, 60 days, 90+ days. Available from the quick links at the bottom of the Sales Ledger page.

Customer Statements

Finance > Sales > Statements > [Customer] generates a statement for a specific customer showing all transactions and running balance.

5. Purchase Ledger

The Purchase Ledger manages supplier invoices, credit notes, and payables. Navigate to Finance > Purchase Invoices.

Purchase Ledger Dashboard

Mirrors the Sales Ledger layout with summary cards for outstanding creditor balances across ageing bands, plus tabs for Invoices, Credit Notes, and Suppliers.

Purchase Invoices

- **New Invoice —** Click New Invoice to create. Select a supplier, enter the supplier's invoice number, add line items with expense account coding, VAT treatment, and optional project/cost centre tags.

Invoice numbering uses the PIN-XXXXXX sequence.

Posting a purchase invoice creates the auto-journal: DR Expense (net), DR VAT Input (VAT), CR Trade Creditors (gross).

Suppliers

Each supplier has a unique code (SUPP-XXXXXX), contact details, and an outstanding balance. Click any supplier row to view their account detail.

Aged Creditors Report

Finance > Purchases > Aged Creditors provides a breakdown of all outstanding purchase invoices grouped by ageing bands.

Supplier Statements

Finance > Purchases > Statements > [Supplier] generates a reconciliation statement to compare against the supplier's own records.

Purchase Orders

Link to purchase orders is available from the Purchase Ledger header. Purchase orders use the PO-XXXXXX sequence and can be matched against purchase invoices for three-way matching.

6. Banking

The Banking module manages bank accounts, transactions, receipts, payments, transfers, and reconciliation. Navigate to Finance > Bank & Payments.

Bank Account Setup

Each bank account record stores: account name, account number, sort code, current balance, currency, and a link to its GL account (nominal code) in the Chart of Accounts.

- **New Account —** Finance > Bank & Payments > New Account. Enter the details and select the GL nominal code to link.

Dashboard

The Bank & Payments page shows:

- Summary cards: Total Balance (across all accounts), Active Accounts count, Unreconciled transaction count
- Bank Account cards — each showing name, sort code/account number, current balance, and reconciliation status
- Recent Transactions table — date, account, description, reference, amount (green for receipts, red for payments), reconciliation status

Receive Payment

Finance > Bank > Receive Payment. Records a customer payment:

- Select the bank account, customer, and amount
- Allocate against one or more outstanding sales invoices
- Auto-journal: DR Bank, CR Trade Debtors (1100)
- Invoice status updates to Partially Paid or Paid depending on remaining balance

Make Payment

Finance > Bank > Make Payment. Records a supplier payment:

- Select the bank account, supplier, and amount
- Allocate against one or more outstanding purchase invoices
- Auto-journal: DR Trade Creditors (2100), CR Bank

Bank Transfer

Finance > Bank > Transfer. Moves funds between bank accounts:

- Select source and destination bank accounts, enter the amount
- Auto-journal: DR Destination Bank GL, CR Source Bank GL

Bank Reconciliation

Finance > Bank > [Account] > Reconcile. The reconciliation screen allows you to match transactions against your bank statement:

- Upload or view the bank statement
- Tick transactions that appear on the statement
- The reconciliation summary shows the statement balance vs book balance and any difference
- Once all items are matched and the difference is zero, mark the reconciliation as complete

> Unreconciled transactions show a yellow warning badge on the bank account card. Fully reconciled accounts show a green tick.

Statement Import

Finance > Bank > [Account] > Import. Upload CSV or OFX bank statement files to automatically create transactions for matching.

Bank Rules

Finance > Bank Rules. Set up automatic categorisation rules for imported transactions. Rules match on description patterns and assign the correct nominal code, VAT treatment, and project tag.

7. Construction Contracts

The Construction Contracts module handles NEC and JCT contract accounting, applications for payment, retention, and CIS deductions. Navigate to Finance > Contracts.

Contract Types

Type	Description
NEC	NEC Engineering and Construction Contract
JCT	JCT Standard Building Contract
Bespoke	Custom contract terms

Contract Lifecycle

Status	Meaning
Draft	Contract created, not yet active
Active	Work in progress, applications can be submitted
Practical Completion	Works substantially complete, defects period begins
Defects Liability	Defects liability period in progress
Final Account	Final account negotiation/agreement stage
Closed	Contract fully completed and closed

Contract Dashboard

The contracts list shows summary cards:

- Active Contracts — number of contracts currently in progress
- Total Contract Value — sum of all current contract values (including variations)
- Certified to Date — total value certified by the contract administrator
- Retention Held — total retention deducted and held
- Outstanding Applications — applications awaiting certification or payment

Creating a Contract

- **New Contract —** Finance > Contracts > New Contract. Enter contract reference, client name, type (NEC/JCT/Bespoke), original value, retention percentage, defects period, and revenue account.

Applications for Payment

Within each contract, you can create applications for payment:

- Finance > Contracts > [Contract] > Applications > New Application
- Enter the gross valuation for the period, cumulative to date, and any materials on site
- The system calculates retention to be held and CIS deductions (if applicable)
- Net payable = Gross valuation - Retention - CIS deductions

Application Certified Auto-Journal

When an application is certified, the auto-journal engine creates:

- DR Trade Debtors (net payable after retention and CIS)
- DR Retention Held (2400) — retention amount
- CR Revenue account — gross valuation
- CR CIS Deductions (2302) — if CIS applies

Retention Management

Finance > Contracts > Retention provides a dedicated view of all retention held across contracts, with release tracking for both first-half and second-half retention.

Variations

Contract variations change the current contract value. The table shows both original value and current value, with the difference highlighted in blue when variations exist.

8. Tax & VAT

The VAT module manages VAT codes, calculates VAT returns, and supports HMRC Making Tax Digital (MTD) submissions. Navigate to Finance > VAT Returns.

VAT Codes

Finance > VAT Returns > Manage Codes (or Finance > VAT > Codes). Each VAT code defines:

- Code identifier (e.g. T0, T1, T2, T5, T9)
- Description (e.g. Zero Rated, Standard Rate, Exempt)
- Rate percentage (e.g. 0%, 20%)
- Whether it appears on VAT returns
- Whether it applies to sales, purchases, or both

VAT Returns

The VAT Returns page shows:

- Summary: Next return due (from periods without returns), last submitted return, outstanding VAT liability
- Returns table: Period, dates, Box 1 (Output VAT), Box 4 (Input VAT), Box 5 (Net VAT), status

VAT Return Statuses

Status	Meaning
VAT_DRAFT	Return created but not yet calculated
CALCULATED	Figures calculated from posted journals in the period
VAT_APPROVED	Reviewed and approved, ready for submission
VAT_SUBMITTED	Submitted to HMRC
ERROR	Submission failed or data issue

Calculating a Return

Click Calculate Return on any period awaiting a VAT return. The system aggregates all posted journal lines tagged with VAT codes in the period and produces the nine-box VAT100 form:

- Box 1 — VAT due on sales and other outputs
- Box 2 — VAT due on acquisitions from EU member states
- Box 3 — Total VAT due (Box 1 + Box 2)
- Box 4 — VAT reclaimed on purchases and other inputs
- Box 5 — Net VAT to pay or reclaim (Box 3 - Box 4)
- Box 6 — Total value of sales excluding VAT
- Box 7 — Total value of purchases excluding VAT
- Box 8 — Total value of supplies to EU member states
- Box 9 — Total value of acquisitions from EU member states

> Box 5 positive = owed to HMRC (shown in red). Box 5 negative = due from HMRC (shown in green).

9. Fixed Assets & Depreciation

The Fixed Assets Register tracks all capitalised assets, manages depreciation calculations, and handles disposals. Navigate to Finance > Fixed Assets.

Asset Register Overview

The landing page shows summary cards: Total Assets count, Total Cost, Total Net Book Value (NBV), and Monthly Depreciation charge.

Asset Record

Each fixed asset record contains:

- Asset code (FA-XXXXXX, auto-generated)
- Name and description
- Category (linked to a depreciation method)
- Purchase date and purchase cost
- Residual / scrap value
- Accumulated depreciation (running total of depreciation charged)
- Net Book Value (cost minus accumulated depreciation)
- Serial number and location (optional)
- Status: Active, Disposed, Fully Depreciated, Written Off

Asset Categories

Finance > Fixed Assets > Categories. Each category defines the depreciation method for all assets assigned to it:

- Straight Line — equal annual charge over the useful life
- Reducing Balance — percentage of the remaining book value each year

Running Depreciation

Finance > Fixed Assets > Run Depreciation. Select a period and the system calculates the depreciation charge for each active asset based on its category method. A depreciation journal is created automatically.

Asset Statuses

Status	Meaning
Active	In use, depreciation runs monthly
Disposed	Sold or scrapped — disposal proceeds and gain/loss recorded
Fully Depreciated	NBV has reached the residual value — no further depreciation
Written Off	Asset removed from register as a write-off

Disposal

When disposing of an asset, record the disposal date, proceeds, and method. The system calculates the gain or loss on disposal (proceeds minus NBV) and creates the appropriate journal entries.

10. Reports

The Reports hub provides all key financial reports. Navigate to Finance > Reports.

Available Reports

Report	Description
Trial Balance	All account balances for a period — verifies debits equal credits
Profit & Loss	Income and expenditure statement for a period or date range
Balance Sheet	Statement of financial position — assets, liabilities, equity
Aged Debtors	Outstanding sales invoices grouped by ageing bands
Aged Creditors	Outstanding purchase invoices grouped by ageing bands
VAT Report	VAT summary with box-by-box breakdown for HMRC
Nominal Activity	Transaction-level detail for any nominal account over a date range
Job Costing	Revenue and cost analysis by project or job code

Trial Balance

The trial balance lists every account with a non-zero balance, showing debit and credit totals. The totals must agree (debits = credits). If they do not, it indicates a posting error that must be investigated.

Profit & Loss

Shows Revenue accounts (credit balances) minus Expense accounts (debit balances) for the selected period. The bottom line is net profit or net loss.

Balance Sheet

Shows the financial position at a point in time: Total Assets = Total Liabilities + Total Equity. This report pulls from Asset, Liability, and Equity type accounts.

Job Costing

Analyses revenue and costs posted against specific projects. Shows gross margin per job, which is critical for MME's project-based work streams (Utility, Bespoke, Community, Blast, Refurbishment).

Nominal Activity

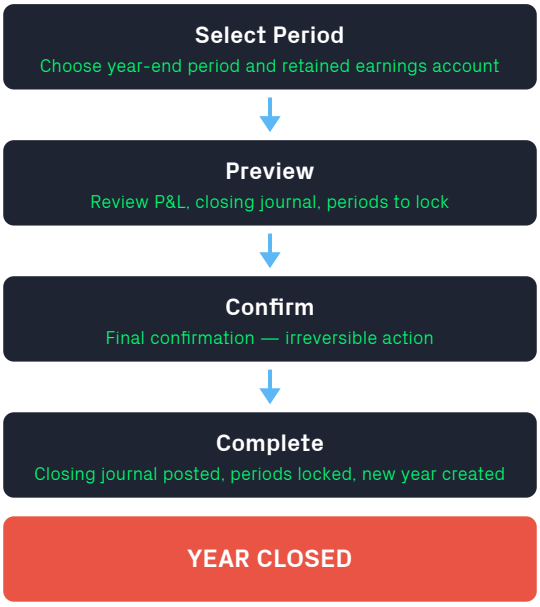
Select any nominal account and a date range to see every journal line posted to that account — useful for drilling down from trial balance variances.

11. Year-End

Year-end processing closes the financial year, transfers the profit or loss to retained earnings, and creates the next year's accounting periods. Navigate to Finance > Year-End.

Year-End Wizard

The year-end process follows a four-step wizard:



Step 1 — Select Period

- Choose the accounting period marked as the year-end period
- Select the retained earnings equity account (usually 3100 Retained Earnings)
- Only periods not already locked are available for selection

Step 2 — Preview

The preview shows:

- P&L summary — total revenue, total expenses, net profit or loss
- Revenue and expense account breakdowns with individual balances
- Closing journal entry preview — the balanced journal that will be posted
- List of periods that will be locked
- Any warnings (e.g. open periods that haven't been closed)

Step 3 — Confirm

Final confirmation screen. This action is irreversible. Shows a summary of what will happen.

Step 4 — Complete

After processing completes, the system:

- Creates and posts the closing journal entry (transfers all P&L balances to Retained Earnings)
- Locks all periods in the financial year — no further postings allowed
- Creates 12 new monthly accounting periods for the next financial year
- Marks the last new period as the next year-end

> Year-end processing is permanent. Ensure all reconciliations, adjustments, and reviews are complete before proceeding.

Accounting Periods

Finance > Periods shows all accounting periods with their status:

Status	Meaning
OPEN	Journals can be posted to this period
PERIOD_CLOSED	Temporarily closed — can be reopened by an authorised user

LOCKED	Permanently locked by year-end processing — cannot be reopened

Close or reopen periods individually from the Periods page. The current period (today's date falls within its range) is highlighted in blue.

Quick Reference — Where to Find Things

Feature	Location
Finance Dashboard	/finance
Chart of Accounts	/finance/chart-of-accounts
Journal Entries	/finance/journals
Sales Ledger	/finance/sales
Credit Control	/finance/credit-control
Aged Debtors	/finance/sales/aged-debtors
Purchase Ledger	/finance/purchases
Aged Creditors	/finance/purchases/aged-creditors
Bank & Payments	/finance/bank
Bank Reconciliation	/finance/bank/accounts/[id]/reconcile
Bank Rules	/finance/bank-rules
Construction Contracts	/finance/contracts
Retention Tracking	/finance/contracts/retention
VAT Returns	/finance/vat
VAT Codes	/finance/vat/codes
Fixed Assets	/finance/fixed-assets
Run Depreciation	/finance/fixed-assets/depreciation
Asset Categories	/finance/fixed-assets/categories
Reports Hub	/finance/reports
Trial Balance	/finance/reports/trial-balance
Profit & Loss	/finance/reports/profit-and-loss
Balance Sheet	/finance/reports/balance-sheet
Job Costing	/finance/reports/job-costing
Nominal Activity	/finance/reports/nominal-activity
Accounting Periods	/finance/periods

Year-End Processing	/finance/year-end
Recurring Entries	/finance/recurring
Prepayments	/finance/prepayments
Cost Centres	/finance/cost-centres
Budgets	/finance/budgets
Sage Export	/finance/exports
Import Data	/finance/import
Finance Settings	/finance/settings